Year 12 Further Maths - Mathematical induction

Similar questions - Divisibility

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Prove by induction that $3^{2n+1} + 2^{n+2}$ is divisible by 7 for all integers $n \geqslant 1$. **[6]**

2 Prove by induction that $7^n - 4^n$ is divisible by 3 for all integers $n \geqslant 1$. **[5]**

3 Prove by induction that $4^n + 6n - 1$ is divisible by 9 for all integers $n \geqslant 1$. **[6]**

Answers

Question 1

Let $f(n) = 3^{2n+1} + 2^{n+2}$. *Base case:* $f(1) = 27 + 8 = 35 = 7 \times 5$. *Assumption:* $f(k) = 7m$ for some integer m. *Step:* $f(k+1) = 3^{2k+3} + 2^{k+3} = 9 \times 3^{2k+1} + 2 \times 2^{k+2} = 2\left(3^{2k+1} + 2^{k+2}\right) + 7 \times 3^{2k+1} = 14m + 7 \times 3^{2k+1}$, divisible by 7. So true for $n = 1$, and if true for $n = k$ then true for $n = k+1$; by induction true for all $n \geqslant 1$

Question 2

Base: $7 - 4 = 3$. *Assume* $7^k - 4^k = 3m$. *Step:* $7^{k+1} - 4^{k+1} = 7 \cdot 7^k - 4 \cdot 4^k = 4\left(7^k - 4^k\right) + 3 \cdot 7^k = 12m + 3 \cdot 7^k$, divisible by 3. Hence true for all $n \geqslant 1$ by induction

Question 3

Base: $4 + 6 - 1 = 9$. *Assume* $4^k + 6k - 1 = 9m$. *Step:* $4^{k+1} + 6(k+1) - 1 = 4 \cdot 4^k + 6k + 5 = 4\left(4^k + 6k - 1\right) - 18k + 9 = 36m - 18k + 9 = 9(4m - 2k + 1)$, divisible by 9. Hence true for all $n \geqslant 1$ by induction